

In information retrieval, tf-idf, short for term frequency-inverse document frequency, is a numerical statistic that is intended to reflect how important a word is to a document in a collection or corpus. It is often used as a weighting factor in searches of information retrieval, text mining, and user modeling. The tf-idf value increases proportionally to the number of times a word appears in the document and is offset by the number of documents in the corpus that contain the word, which helps to adjust for the fact that some words appear more frequently in general.

The term frequency of a term in a document is the number of times that the term occurs in that document. Raw term frequency is rarely used directly, because relevance does not increase proportionally with term frequency: a document mentioning a term ten times is more relevant than one mentioning it once, but not ten times more relevant. For this reason the log frequency weight is commonly used, defined as one plus the logarithm of the term frequency when the term occurs, and zero otherwise.

The inverse document frequency is a measure of how much information a term provides, that is, whether it is common or rare across all documents. It is obtained by dividing the total number of documents by the number of documents containing the term, and then taking the logarithm of that quotient. Rare terms such as technical vocabulary receive high weights, while terms that appear in most documents receive weights near zero.

Document length normalization addresses a further problem: longer documents contain more terms and would otherwise be retrieved more often regardless of relevance. Cosine normalization divides each document vector by its Euclidean length, so that similarity depends on the direction of the vector rather than its magnitude. The combination of log term frequency, inverse document frequency, and cosine normalization is written in SMART notation, where the scheme Inc.ltc is a widely used standard.